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Section-6

OSSP (Operating Systems And Systems Programming) - 25CS2104E

(SKILL Content Delivery Concepts)

13.To Parse Input Redirection, Open Files, Redirect stdin, Handle Missing Files, Restore Streams, Test File Inputs.

To Parse Output Redirection, Create Output Files, Redirect stdout, Handle Permissions, Verify Output Data, Test Redirection.

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~$ ls
Linux-System-Programming-Week1  OSSP-Practical      foldername  moni    sample1.c
OSLAB                          Simple-Commands-Week1 hello       monis   sample1.c.save
OSLAB14                        example.txt         hello.c     sample1
monis@DESKTOP-C3GM4T6:~$ cd OSLAB
monis@DESKTOP-C3GM4T6:~/OSLAB$ ls
OSLAB28  Scheduling-Algorithms  hello  hello.c:Zone.Identifier  hello_link  newfile.txt  test1
Pipes    file1.txt              hello.c  hello:Zone.Identifier    hello_shortcut skill
monis@DESKTOP-C3GM4T6:~/OSLAB$ cd skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 2 — Create the Input Redirection Program

```
Select monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano input_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 input_redirection.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define INPUT_SIZE 200

int main()
{
    char input[INPUT_SIZE];
    int saved_stdin;
    int fd;

    printf("Input Redirection Program\n");
    printf("Enter a command in the form: cat < filename\n");
    printf("Type exit to quit.\n");

    while (1)
    {
        printf("redirect> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        char *redirect = strchr(input, '<');

        if (redirect == NULL)
        {
            printf("Error: Input redirection '<' not found.\n");
            continue;
        }

        *redirect = '\0';

        char *filename = redirect + 1;

        while (*filename == ' ')
            filename++;

        ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
    }
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 input_redirection.c *
    filename++;

    if (*filename == '\\0')
    {
        printf("Error: Input file name is missing.\n");
        continue;
    }

    fd = open(filename, O_RDONLY);

    if (fd == -1)
    {
        perror("open");
        printf("Error: Cannot open input file: %s\n", filename);
        continue;
    }

    saved_stdin = dup(STDIN_FILENO);

    if (saved_stdin == -1)
    {
        perror("dup");
        close(fd);
        continue;
    }

    if (dup2(fd, STDIN_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        close(saved_stdin);
        continue;
    }

    close(fd);

    printf("Input redirected from: %s\n", filename);

    char buffer[100];

    while (fgets(buffer, sizeof(buffer), stdin) != NULL)
    {
        printf("Read: %s", buffer);
    }

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^/ Go To Line M-E Redo
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 input_redirection.c *
}

saved_stdin = dup(STDIN_FILENO);

if (saved_stdin == -1)
{
    perror("dup");
    close(fd);
    continue;
}

if (dup2(fd, STDIN_FILENO) == -1)
{
    perror("dup2");
    close(fd);
    close(saved_stdin);
    continue;
}

close(fd);

printf("Input redirected from: %s\n", filename);

char buffer[100];

while (fgets(buffer, sizeof(buffer), stdin) != NULL)
{
    printf("Read: %s", buffer);
}

if (dup2(saved_stdin, STDIN_FILENO) == -1)
{
    perror("dup2 restore");
}

close(saved_stdin);

printf("\nInput stream restored successfully.\n");
}

printf("Program exited successfully.\n");

return 0;
}
```

^G Help ^O Write Out ^F Where Is ^K Cut ^T Execute ^C Location M-U Undo
^X Exit ^R Read File ^\ Replace ^U Paste ^J Justify ^_ Go To Line M-E Redo

Save in nano:

1. Press **Ctrl + O**
2. Press **Enter**
3. Press **Ctrl + X**

```
Write to File: input_redirection.c
^G Help   M-D DOS Format   M-A Append   M-B Backup File   ^T Browse
^C Cancel   M-M Mac Format   M-P Prepend   ^Q Discard buffer
```

Then compile it:

`gcc input_redirection.c -o input_redirection`

Check that compilation succeeded:

`ls -l input_redirection.c input_redirection`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano input_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc input_redirection.c -o input_redirection
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l input_redirection.c input_redirection
-rwxr-xr-x 1 monis monis 16568 Sep 18 11:32 input_redirection
-rw-r--r-- 1 monis monis  2067 Sep 18 11:32 input_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 3 — Create a Test Input File

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano input.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 input.txt *
Operating Systems and Systems Programming
Input redirection test
Data read from a file using stdin
```

Save:

1. **Ctrl + O**
2. **Enter**
3. **Ctrl + X**

```
Write to File: input.txt
^G Help      M-D DOS Format  M-A Append     M-B Backup File ^T Browse
^C Cancel    M-M Mac Format  M-P Prepend    ^O Discard buffer
```

Now verify the contents:

cat input.txt

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano input.txt
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat input.txt
Operating Systems and Systems Programming
Input redirection test
Data read from a file using stdin
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 4 — Test Input Redirection

Now run the program:

`./input_redirection`

```
Select monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./input_redirection
Input Redirection Program
Enter a command in the form: cat < filename
Type exit to quit.
redirect>
```

At the prompt, enter:

`cat < input.txt`

```
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./input_redirection
Input Redirection Program
Enter a command in the form: cat < filename
Type exit to quit.
redirect> cat < input.txt
Input redirected from: input.txt
Read: Operating Systems and Systems Programming
Read: Input redirection test
Read: Data read from a file using stdin

Input stream restored successfully.
redirect> Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Test missing file

At the next prompt, enter:

`cat < missing.txt`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./input_redirection
Input Redirection Program
Enter a command in the form: cat < filename
Type exit to quit.
redirect> cat < input.txt
Input redirected from: input.txt
Read: Operating Systems and Systems Programming
Read: Input redirection test
Read: Data read from a file using stdin

Input stream restored successfully.
redirect> Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./input_redirection
Input Redirection Program
Enter a command in the form: cat < filename
Type exit to quit.
redirect> cat < missing.txt
open: No such file or directory
Error: Cannot open input file: missing.txt
redirect>
```

Exit

Finally enter:

exit

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./input_redirection
Input Redirection Program
Enter a command in the form: cat < filename
Type exit to quit.
redirect> cat < input.txt
Input redirected from: input.txt
Read: Operating Systems and Systems Programming
Read: Input redirection test
Read: Data read from a file using stdin

Input stream restored successfully.
redirect> Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./input_redirection
Input Redirection Program
Enter a command in the form: cat < filename
Type exit to quit.
redirect> cat < missing.txt
open: No such file or directory
Error: Cannot open input file: missing.txt
redirect> exit
Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 5 — Create the Output Redirection Program

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano output_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

```
GNU nano 8.7.1 output_redirection.c *
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <fcntl.h>
#include <unistd.h>

#define INPUT_SIZE 200

int main()
{
    char input[INPUT_SIZE];
    int saved_stdout;
    int fd;

    printf("Output Redirection Program\n");
    printf("Enter a command in the form: echo Hello > output.txt\n");
    printf("Type exit to quit.\n");

    while (1)
    {
        printf("redirect> ");
        fflush(stdout);

        if (fgets(input, sizeof(input), stdin) == NULL)
            break;

        input[strcspn(input, "\n")] = '\0';

        if (strcmp(input, "exit") == 0)
            break;

        char *redirect = strchr(input, '>');

        if (redirect == NULL)
        {
            printf("Error: Output redirection '>' not found.\n");
            continue;
        }

        *redirect = '\0';

        char *command = input;
        char *filename = redirect + 1;

        while (*filename == ' ')
            filename++;

        ^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
        ^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line  M-E Redo
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 output_redirection.c *
while (*filename == ' ')
    filename++;

if (*filename == '\0')
{
    printf("Error: Output file name is missing.\n");
    continue;
}

while (*command == ' ')
    command++;

if (*command == '\0')
{
    printf("Error: Command is missing.\n");
    continue;
}

fd = open(filename, O_WRONLY | O_CREAT | O_TRUNC, 0644);

if (fd == -1)
{
    perror("open");
    printf("Error: Cannot open output file: %s\n", filename);
    continue;
}

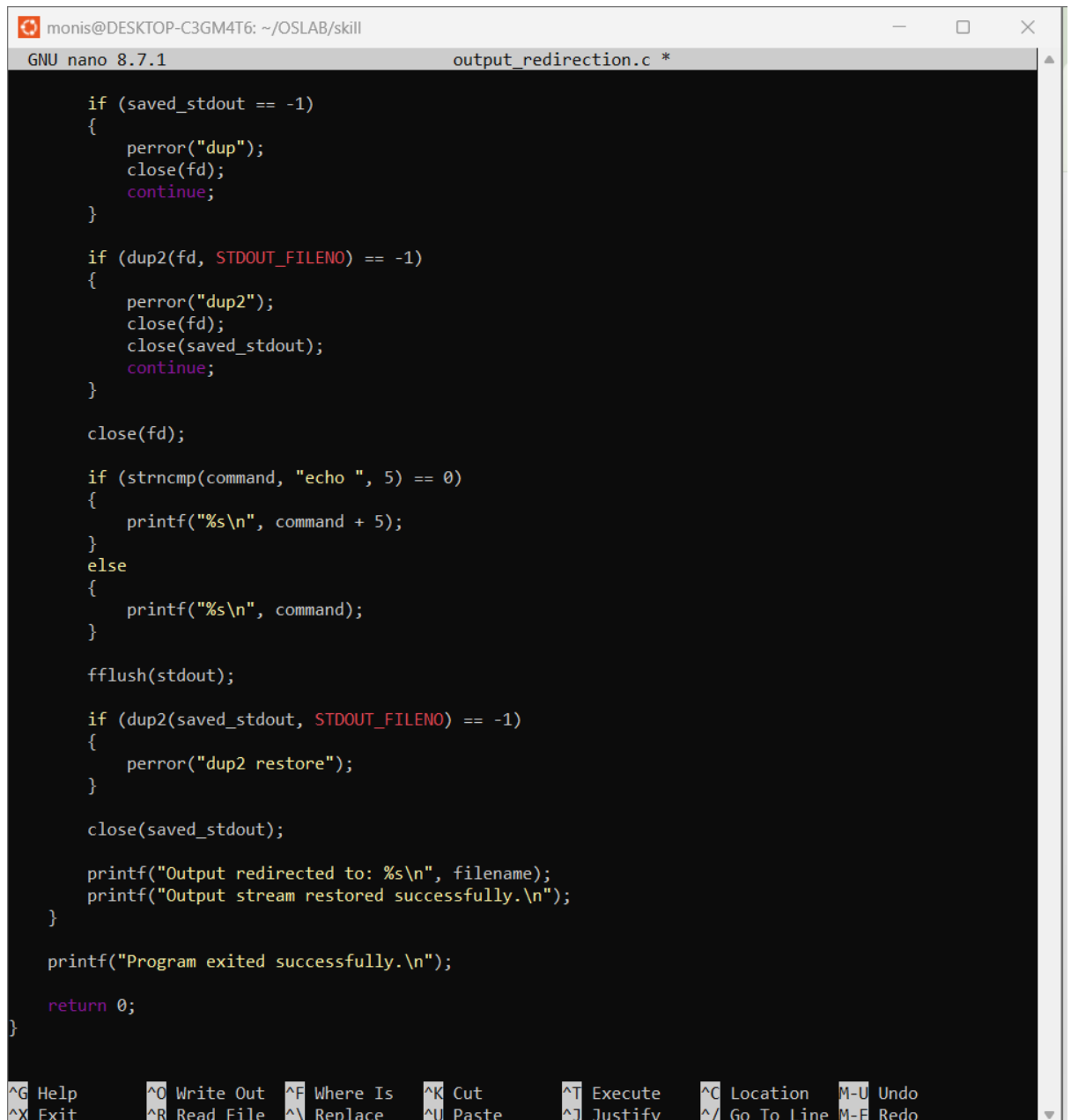
saved_stdout = dup(STDOUT_FILENO);

if (saved_stdout == -1)
{
    perror("dup");
    close(fd);
    continue;
}

if (dup2(fd, STDOUT_FILENO) == -1)
{
    perror("dup2");
    close(fd);
    close(saved_stdout);
    continue;
}

close(fd);

^G Help      ^O Write Out  ^F Where Is   ^K Cut        ^T Execute    ^C Location   M-U Undo
^X Exit      ^R Read File  ^\ Replace    ^U Paste      ^J Justify    ^_ Go To Line M-E Redo
```



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
GNU nano 8.7.1 output_redirection.c *

    if (saved_stdout == -1)
    {
        perror("dup");
        close(fd);
        continue;
    }

    if (dup2(fd, STDOUT_FILENO) == -1)
    {
        perror("dup2");
        close(fd);
        close(saved_stdout);
        continue;
    }

    close(fd);

    if (strncmp(command, "echo ", 5) == 0)
    {
        printf("%s\n", command + 5);
    }
    else
    {
        printf("%s\n", command);
    }

    fflush(stdout);

    if (dup2(saved_stdout, STDOUT_FILENO) == -1)
    {
        perror("dup2 restore");
    }

    close(saved_stdout);

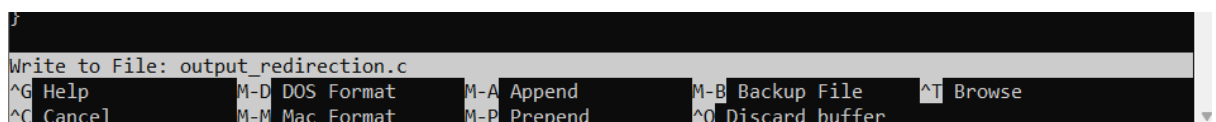
    printf("Output redirected to: %s\n", filename);
    printf("Output stream restored successfully.\n");
}

printf("Program exited successfully.\n");

return 0;
}
```

Save:

- **Ctrl + O**
- **Enter**
- **Ctrl + X**



Compile:

gcc output_redirection.c -o output_redirection

Verify:

ls -l output_redirection.c output_redirection

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ nano output_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ gcc output_redirection.c -o output_redirection
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ls -l output_redirection.c output_redirection
-rwxr-xr-x 1 monis monis 16608 Sep 18 11:49 output_redirection
-rw-r--r-- 1 monis monis 2379 Sep 18 11:48 output_redirection.c
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 6 — Test Output Redirection

Run the program:

`./output_redirection`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./output_redirection
Output Redirection Program
Enter a command in the form: echo Hello > output.txt
Type exit to quit.
redirect>
```

At the prompt, enter:

`echo Hello from output redirection > output.txt`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./output_redirection
Output Redirection Program
Enter a command in the form: echo Hello > output.txt
Type exit to quit.
redirect> echo Hello from output redirection > output.txt
Output redirected to: output.txt
Output stream restored successfully.
redirect>
```

Now verify the file:

`cat output.txt`

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat output.txt
Hello from output redirection
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Test that the file can be overwritten

Run the program again:

./output_redirection

Enter:

echo Second output test > output.txt

Then:

exit

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./output_redirection
Output Redirection Program
Enter a command in the form: echo Hello > output.txt
Type exit to quit.
redirect> echo Second output test > output.txt
Output redirected to: output.txt
Output stream restored successfully.
redirect>
```

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./output_redirection
Output Redirection Program
Enter a command in the form: echo Hello > output.txt
Type exit to quit.
redirect> echo Second output test > output.txt
Output redirected to: output.txt
Output stream restored successfully.
redirect> exit
Program exited successfully.
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Verify:

cat output.txt

```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat output.txt
Second output test
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```

Step 7 — Test Missing Output File / Permission Handling

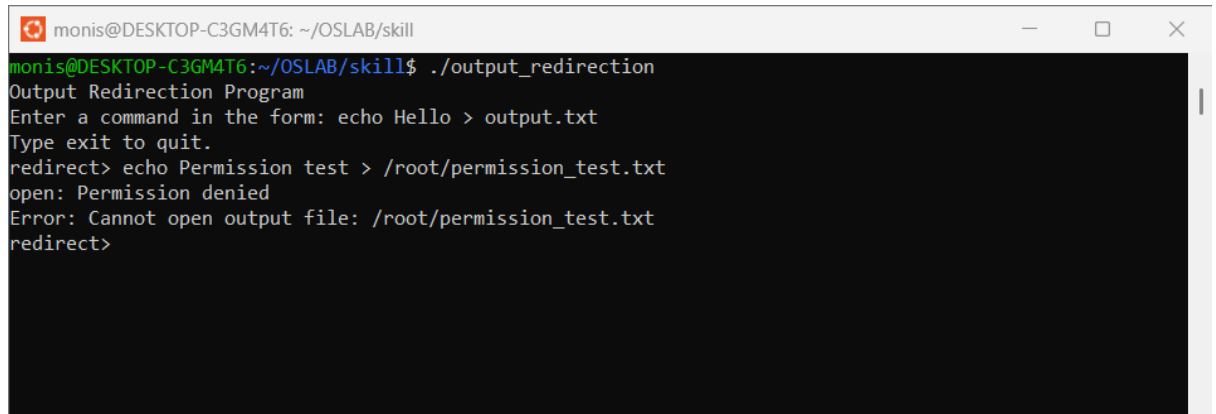
Now let's test the error-handling part of output redirection.

Run:

./output_redirection

At the prompt, enter:

echo Permission test > /root/permission_test.txt

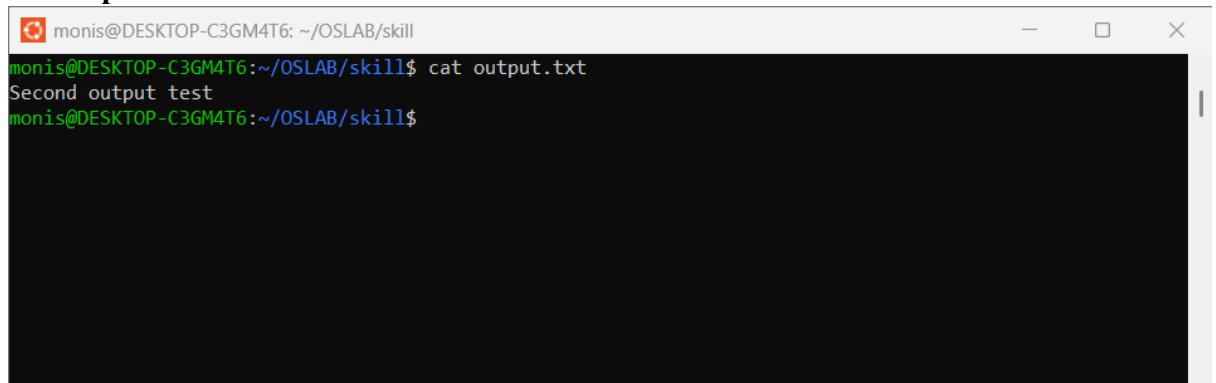


```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ ./output_redirection
Output Redirection Program
Enter a command in the form: echo Hello > output.txt
Type exit to quit.
redirect> echo Permission test > /root/permission_test.txt
open: Permission denied
Error: Cannot open output file: /root/permission_test.txt
redirect>
```

Verify your normal output file still exists

Run:

cat output.txt



```
monis@DESKTOP-C3GM4T6: ~/OSLAB/skill
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$ cat output.txt
Second output test
monis@DESKTOP-C3GM4T6:~/OSLAB/skill$
```